

1

- (a) The database needs three tables to store the data for the students' enrolment for the courses: STUDENT, ENROLMENT and COURSE.

Draw an Entity-Relationship (E-R) diagram showing the three tables and the relationships between them. [2]

	STUDENT one-to-many ENROLMENT
	ENROLMENT many-to-one COURSE

(b)

Draw an Entity-Relationship (E-R) diagram showing the four tables and the relationships between them. [3]

	COORDINATOR one-to-many COURSE
	COURSE one-to-many ENROLMENT
	ENROLMENT one-to-one PROJECT

- (c) Draw the overall Entity-Relationship (E-R) diagram showing the above tables and the relationships between them. [1]

	Combine the above 4 tables correctly
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(d)

The ENROLMENT table description has been done for you. Based on the above information, write the table descriptions for the other four tables, each of which are in third normal form (3NF). [4]

ENROLMENT (StudentID, CourseCode, DateEnrolled, FinalGrade)

The primary key is indicated by underlining one or more attributes. The foreign key is indicated by drawing a dashed line below one or more attributes.

	STUDENT (StudentID (PK), Name, ContactNo, EmailAddr)
	COORDINATOR (CoordinatorID (PK), Name, EmailAddr)
	COURSE (CourseCode (PK), Name, CoordinatorID (FK))
	PROJECT (StudentID (PK,FK), CourseCode (PK,FK), ProjectID (PK), Grade)

- (e) Singapore's Personal Data Protection Act (PDPA) protects data of all students stored in the database. Describe one way in which PDPA achieves this purpose. [2]

	State the obligation
	Explain the obligation https://www.pdpc.gov.sg/Overview-of-PDPA/The-Legislation/Personal-Data-Protection-Act/Data-Protection-Obligations

2 Cyberattacks have become more prevalent due to the advent of technology.

- (a) With the aid of a diagram, explain how a denial of service (DOS) attack can disrupt a server's operations. [2]

	Correct simplified diagram: Attacker's computer ->router->Victim's server.
	Any one of the following: Vulnerability attack This involves sending a few well-crafted messages to a vulnerable application or operating system running on a targeted host. If the right sequence of packets is sent to a vulnerable application or operating system, the service can stop or, worse, the host can crash. Bandwidth flooding The attacker sends a deluge of packets to the targeted host—so many packets that the target's access link becomes clogged, preventing legitimate packets from reaching the server. Connection flooding The attacker establishes a large number of half-open or fully open TCP connections at the target host. The host can become so bogged down with these bogus connections that it stops accepting legitimate connections.

- (b) State another malware that can also disrupt a server's operations and explain how it does so. [2]

	A worm is a standalone computer program that runs automatically and attempt to spread copies of itself over a network.
	Worms consume bandwidth and overload web servers, harming the host network.
	Any other logical answer

- (c) Network operational devices such as firewalls, intrusion detection system (IDS), and intrusion prevention system (IPS) can be used to counter attacks against a network.

Explain how a firewall is deployed to block unwanted traffic from accessing a particular network. [2]

	All traffic from outside to inside, and vice versa, passes through the firewall, which sits squarely at the boundary between the administered network and the rest of the Internet.
	Only authorised traffic, as defined by the local security policy, will be allowed to pass.

- (d) Often, intrusion detection system (IDS) and intrusion prevention system (IPS) collectively known as IDS systems are used in tandem with firewall for better detection of network attacks. Explain how IDS and IPS help in providing an additional layer of protection for a network. [2]

	IDS is a device that generates alerts when it observes potentially malicious traffic. IPS is a device that filters out suspicious traffic.
	To detect many attack types, these IDS systems perform deep packet inspection, that is, look beyond the header fields and into the actual application data that the packets carry.

3

- (a) Explain what is meant by a communication protocol and how it is used to handle the complexity in transmitting data over the Internet. [2]

	A protocol is a set of rules for data transmission which are agreed upon by both sender and receiver.
	To handle complexity in data communication, we can break the system down and organise its components into layers where each layer provides functionality that depends solely on the layer directly below it.

- (b) State the name of the protocol that is used to access the different web pages on the Internet. Briefly describe how this protocol works. [3]

1m	Hypertext Transfer Protocol (HTTP)
2m	The client (usually a browser) opens a connection to the server via the TCP/IP protocol and sends a HTTP request.
	The server receives this HTTP request, interprets it and respond with the appropriate HTTP response.
	The client receives this HTTP response, interprets it and displays it in the browser's window.

- (c) Explain why a client-server architecture is implemented in the case of a web application. [2]

	Any 2 of the following or any other logical explanation.
	Need an always-on Web server services requests from many browsers running on client hosts simultaneously.
	Client and server have distinct tasks. Client sends request while server responds by sending requested object to the client.
	The clients, i.e. the web browsers in this case, do not directly communicate with each other. Therefore, there is a need to maintain consistency across all devices, i.e. not platform-specific.

- (d) Describe how Domain Name System (DNS) allows access to a website using a Uniform Resource Locator (URL). [3]

	A URL is a complete web address used to find a particular web page. The host's domain name is part of this URL. When a client requests access to the website, the DNS system will need to contact the host's DNS server.
	This DNS server can be reached since the host's Internet registrar has placed an entry for the host's DNS server (domain name and corresponding IP address) in the top-level domain (TLD) servers. In other words, the DNS system maps the domain names with their corresponding IP addresses.
	Upon contacting the DNS server, the client will receive the IP address of the host's Web server and the client will then be able to access the website.
*	Students may also include the use of cache and recursive name servers to search for DNS record. If it is not found then, it will contact the TLD servers.

4

- (a) Discuss whether iterative linear search is better than recursive linear search. [2]

	Iterative linear search would be better than recursive linear search as recursive linear search will need a larger amount of memory space to store the return addresses and copies of local and temporary variables from the multiple function calls whereas iterative linear search doesn't.
	If a large data list is involved, executing a recursive linear search may cause the processor to run out of memory and the program to crash.

(b)

- (i) Given the array [9, 12, 9, 2, 17, 1, 6], trace how the items are sorted using quicksort. [4]

	[2, 1, 6] [9] [12, 9, 17]
	[1][2][6] [9] [9][12][17]
	[1, 2, 6] [9] [9, 12, 17]
	[1, 2, 6, 9, 9, 12, 17]

- (ii) Evaluate and justify the circumstances when quicksort should be used and shouldn't be used as the sorting algorithm. [2]

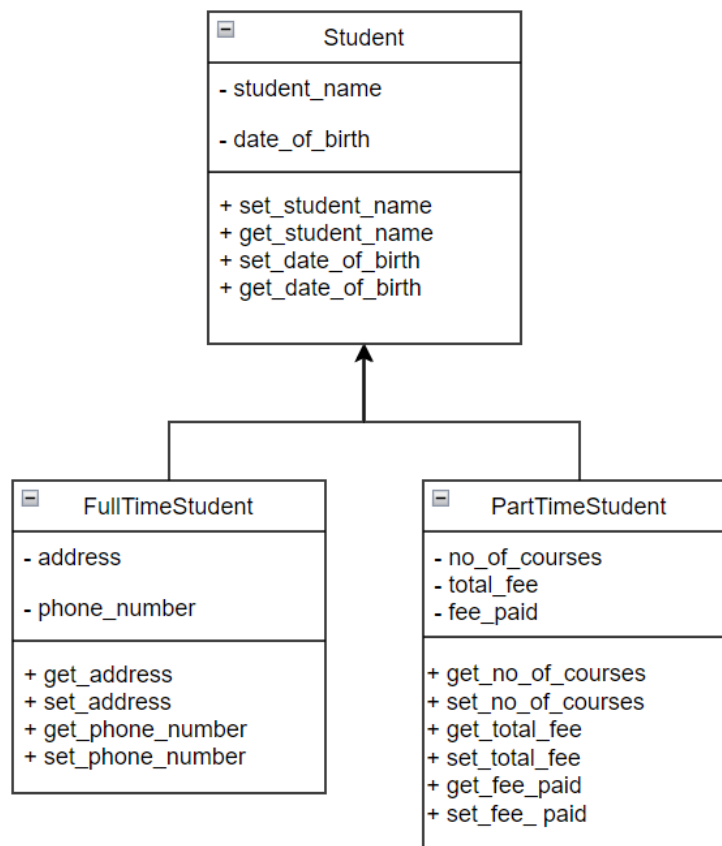
	Quicksort should not be used to sort list that is largely sorted as the pivot chosen might end up to be always the smallest or largest element and cause the subarrays to be unequally divided which may cause a worse case time complexity of $O(n^2)$
	Quicksort is an unstable sort. It should be used to sort list that does not contain duplicate values or items so as to maintain the sequence of occurrence of the items in the list.

5

(a) Draw a class diagram that shows the following for the situation described above:

- the superclass
- any subclass
- inheritance
- properties
- appropriate methods

[7]



(b) State the purpose of encapsulation.

[2]

1	Ensuring private/internal properties are only accessed/alterd by calls to exposed/public methods
1	Have private data/properties and public methods together in one entity called a class.

A part-time student can be either a local student or a foreign student. A foreign student will need to pay an additional administration fee for each enrolled course.

(c) Suggest a change to the class diagram to enable the new calculation of total fee used for foreign part-time student.

[1]

1	Create a subclass ForeignStudent, with PartTimeStudent as parent class
1	Re-define set_total_fee in ForeignStudent to override set_total_fee method in PartTimeStudent.

6

(a) Create a decision table to show these conditions and actions.

[4]

Conditions	Document printed but quality is poor
	Error light flashing on printer
	Document printed but paper size is incorrect

		Rules							
Conditions	Document printed but quality is poor	Y	Y	Y	Y	N	N	N	N
	Error light flashing on printer	Y	Y	N	N	Y	Y	N	N
	Document printed but paper size is incorrect	Y	N	Y	N	Y	N	Y	N
Actions	Check connection from computer to printer						X		
	Check ink status	X	X	X	X				
	Check whether there is paper jam						X		
	Check paper size	X		X		X		X	

(b) Simplify your decision table by removing redundancies from the decision table.

[2]

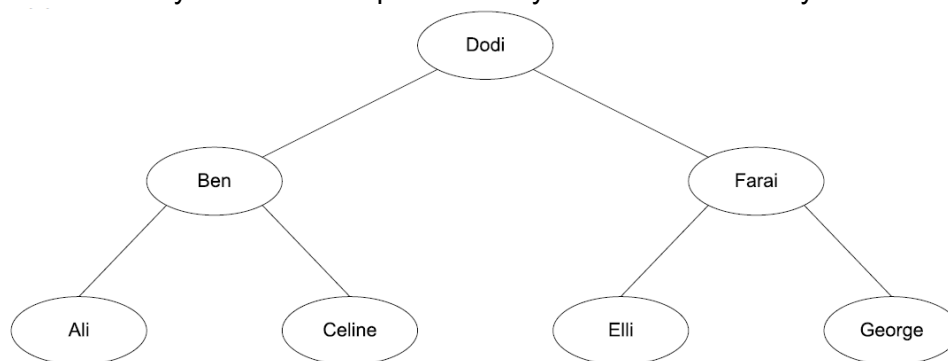
		Rules							
Conditions	Document printed but quality is poor	Y	Y	N	N	N			
	Error light flashing on printer	-	-	-	Y	N			
	Document printed but paper size is incorrect	Y	N	Y	N	N			
Actions	Check connection from computer to printer				X				
	Check ink status	X	X						
	Check whether there is paper jam				X				
	Check paper size	X		X					

- 7 A binary search tree is implemented using an array, `Names`. The names are stored in the elements of the array in alphabetical order. Each element of the array `Names` comprises three parts: a left pointer, the data and a right pointer.

Index	<code>l_ptr</code>	data	<code>r_ptr</code>
0	-1	Celine	-1
1	-1	Elli	-1
2	3	Ben	0
3	-1	Ali	-1
4	2	Dodi	6
5	-1	None	8
6	1	Farai	7
7	-1	George	-1
8	5	None	-1
root			
4			
free_pointer			
5			

- (a) Draw the binary search tree represented by the data in the array `Names`.

[2]



- (b) A new name, **Faith**, is to be inserted into the binary search tree. Show the changes to the array `Names` after this insertion.

[3]

Index	<code>l_ptr</code>	data	<code>r_ptr</code>
0	-1	Celine	-1
1	-1	Elli	5
2	3	Ben	0
3	-1	Ali	-1
4	2	Dodi	6
5	-1	Faith	-1
6	1	Farai	7
7	-1	George	-1
8	-1	None	-1

(c) State the index of `free_pointer` after the insertion in part (b).

[1]

	8

(d) A programmer needs an algorithm to output the result of an in-order traversal of Names. To design this, the programmer writes a recursive procedure in pseudocode.

```

01 PROCEDURE TraverseTree(BYVALUE Root: INTEGER)
02   IF Tree[Root].LeftPointer .....A.....
03   THEN
04       TraverseTree(.....B.....)
05   ENDIF
06   OUTPUT .....C..... .Name
07   IF .....D..... <> -1
08   THEN
09       TraverseTree(.....E.....)
10   ENDIF
11 ENDPROCEDURE

```

Complete the pseudocode for A, B, C, D and E in the algorithm.

[5]

	A - <> 0
	B - Tree[Root].LeftPointer
	C - Tree[Root]
	D - Tree[Root].RightPointer
	E - Tree[Root].RightPointer

```

01 PROCEDURE TraverseTree(BYVALUE Root : INTEGER)
02   IF Tree[Root].LeftPointer < > 0
03   THEN
04       TraverseTree(Tree[Root].LeftPointer)
05   ENDIF
06   OUTPUT Tree[Root].Name
07   IF Tree[Root].RightPointer < > 0
08   THEN
09       TraverseTree(Tree[Root].RightPointer)
10   ENDIF
11 ENDPROCEDURE

```